Sahaj Bhadja

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Summary

Software Engineer graduating May 2026 with 2+ years building distributed services, scalable APIs, and full-stack platforms using TypeScript, JavaScript, Java, Python, microservices, Kubernetes, Docker, cloud infrastructure, CI/CD, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Distributed Systems, Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected multi-tenant services on Azure SQL and Blob Storage with row-level isolation for 500+ users, reducing indexed-query latency from 1.2 seconds to 350 milliseconds through schema and access optimization.
- Restructured fault-tolerant asynchronous orchestration using FastAPI workers, processing 1K+ jobs with explicit state transitions and failure isolation while tripling throughput under concurrent workloads and preventing cascading failures.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent backend workflows with multithreading and secure ingestion, processing 1K+ files per batch through AWS RDS and REST APIs with isolated, observable, and recoverable execution under load.
- Implemented production services handling 10K+ daily requests for 150+ users, applying automated tests and CI/CD deployment workflows to reduce turnaround 60% while maintaining stable peak-load API performance.

Aasma Technology Solutions **August 2023 – July 2024**
Software Development Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments consistently.
- Optimized object-oriented API integrations with AWS Glue and S3, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight MERN microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, supporting 2M+ synchronized records through REST APIs and real-time WebSocket updates across production workflows.
- Automated application deployments with Jenkins CI/CD and Docker packaging, achieving 95%+ successful build validation and cutting cloud release time by 50% through repeatable environment configuration across deployments.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Flask APIs for cross-platform booking workflows covering reservations, time slots, pricing, coupons, payments, and status updates while synchronizing 50K+ records through Firebase with sub-200ms responses consistently.
- Designed end-to-end system architecture, mapped 12+ backend integration points, and presented technical tradeoffs to client stakeholders before implementation, reducing design errors 25% and strengthening early delivery alignment.

Skills

Languages: TypeScript, JavaScript, Java, Python, C++, C#, SQL, HTML, CSS, Bash
Platform & Backend: REST APIs, Microservices, Distributed Systems, Node.js, FastAPI, Flask, Spring Boot, WebSockets, OOP
Cloud & Data: AWS, Azure, GCP, Kubernetes, Docker, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Firebase
Delivery & Quality: GitHub, CI/CD, Jenkins, GitHub Actions, Maven, PyTest, Mockito, Unit Testing, Integration Testing, Code Reviews, Agile

Projects

Chat2SQL Code | *Java, TypeScript, Node.js, Docker, Maven*

- Extended an AI-powered database workspace across Java and TypeScript services, supporting 10+ databases with schema exploration, validated query generation, Docker deployment, and cross-platform development through Maven, Node.js, and Yarn.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a scalable fintech platform supporting 50K+ users and 1M+ transactions with secure bank integrations, token-based request controls, real-time synchronization, advanced filtering, and reliable Kubernetes-backed deployment workflows.

Big-Brain Live | *PostgreSQL, REST APIs, GCP, Convex*

- Engineered cloud ingestion and indexing pipelines processing 50K+ records, implementing incremental updates, metadata persistence, authentication, real-time collaboration, and performance tuning that consistently reduced retrieval latency by 40%.